

Jucheng Shen

jucheng.shen@rice.edu | [juchengshen.github.io](https://github.com/juchengshen) | github.com/juchengshen

EDUCATION

Rice University

Expected May 2027

B.A. in Computer Science, Mathematics, and Mathematical Economic Analysis, GPA – 3.8/4.0

- Relevant Coursework: Reinforcement Learning, Computer Vision, Honors Linear Algebra, Honors ODE, Real Analysis, Differential Geometry, Advanced Algorithms and Data Structures, Concurrent Programming

EXPERIENCE

Research Intern | California Institute of Technology, advised by Prof. Anima Anandkumar

May 2026 – Present

- Leading a project that explores learning continuous language representations for diffusion LLMs through spectral decomposition and control.
- Training diffusion LLMs on representations altered by spectral decomposition and control, and evaluating their performance on generalization, language modeling, translation, and summarization.

Research Intern | Rice University, advised by Prof. Anastasios Kyrillidis

Jan 2026 – Sep 2026

- Proposed *Asymmetric Input Recurrence (AIR)*, a shared-weight recurrent Transformer that induces proposal-like z_H and scratchpad-like z_L roles through asymmetric input injection.
- Designed decoded-state, input-ablation, freeze, and attention analyses showing local z_L updates, global z_H updates, and AIR variants that match or exceed HRM with half the parameters.

Research Intern | Princeton University, advised by Prof. Zhuang Liu

Nov 2025 – Feb 2026

- Built the dataset selection and filtering pipeline for *i1*, a fully open text-to-image model trained from scratch.
- Ran 300+ ablations across data, architecture, and training on a project using over 700,000 TPU v6e hours.

Research Intern | UT Austin & Intel Labs, advised by Prof. Atlas Wang & Dr. Souvik Kundu

May 2025 – Sep 2025

- Proposed *One-Shot Dynamic Thresholding (OSDT)*, a training-free adaptive decoding method for diffusion LLMs achieving 24–50% higher throughput without accuracy loss.
- Led a collaboration with Intel Labs, proposing *CadLLM*, a unified adaptive decoding system that jointly tunes step size, block size, vocabulary, and thresholds, achieving up to $2.3\times$ speedup on diffusion LLMs.

Research Intern | Rice University, advised by Prof. Yuke Wang

Apr 2025 – Aug 2025

- Implemented *SuperGen* on two baselines: generate a low-resolution video, upscale it, then refine shifted tiles so a 2–4K clip need not fit on one GPU.
- Conducted end-to-end and ablation experiments, assisted with figure production and paper writing for publication.

PUBLICATIONS

One Model, Two Roles: Emergent Specialization in a Shared Recurrent Transformer. [NeurIPS 2026](#). **Shen, J.***, Su, W.*, Kyrillidis, A. (*equal contribution)

i1: A Simple and Fully Open Recipe for Strong Text-to-Image Models. [arXiv 2026](#). Zeng, B., Luo, T., Pu, S., **Shen, J.**, Lu, T., Sarch, G. H., Liu, Z.

Improving the Throughput of Diffusion-based Large Language Models via a Training-Free Confidence-Aware Calibration. [Findings of ACL 2026](#). **Shen, J.**, Sarkar, G., Ro, Y., Nittur Sridhar, S., Wang, Z., Akella, A., Kundu, S.

Beyond Static Cutoffs: One-Shot Dynamic Thresholding for Diffusion Language Models. [NeurIPS 2025 Efficient Reasoning Workshop](#). **Shen, J.**, Ro, Y.

SuperGen: An Efficient Ultra-High-Resolution Video Generation System with Sketching and Tiling. [arXiv 2025](#). Ye, F., Zhao, Z., Mu, Y., **Shen, J.**, Li, R., Wang, K., Sun, D., Agarwal, S., Lee, M., Cao, T., Akella, A., Krishnamurthy, A., Ng, T. S. E., Tu, Z., Wang, Y.

PROJECTS

Board Member (Tracks and Workshops) | HackRice

Apr 2025 – Oct 2025

- Hosted 2 technical workshops for 100+ attendees; ran Q&A and live coding demos.
- Co-designed and shipped the official starter code used by 500+ hackers ([repo](#)).
- Designed competition tracks and judging criteria for HackRice 15.

HONORS & AWARDS

Frank Liu Jr. Prize for Creative Innovations in Music, Fashion, and the Arts in Napier Rice Launch Challenge ([link](#)).